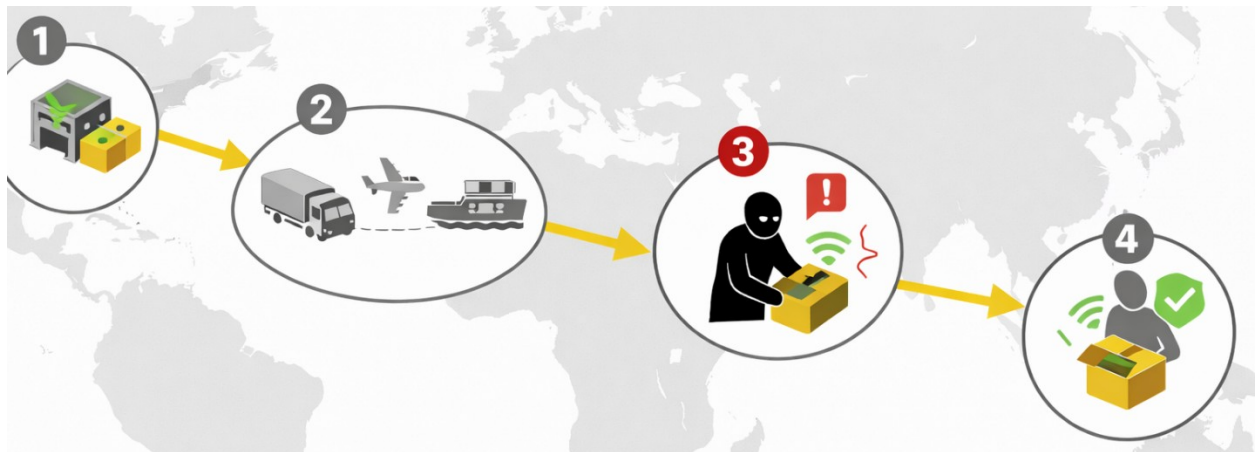


Endpoint Assurance

MANTIS

Multilayered Anti-tamper Network for Trusted Integrated Systems

15.362 Engineering Innovation: Global Security Systems
© Massachusetts Institute of Technology 2026



Team members:

Garrett Campagna, Josh Cohen, Corey Decker, Fritz Demopolous, Mike Fischer, Amit Govrin, Cam Heard, Evan Hoch, Aidan Kenealy, Josephine Lee, Brendan Owen, Greta Reitenbach, Sydney Saenz, Kenan Sehnawi, Chandler Smith, Alfred Wong, Sam Yallowitz

1 Abstract

This paper addresses the problem of discrete parcel tampering in complex, multi-tier logistics networks, where shipments frequently pass through numerous intermediaries and existing detection methods fail to provide reliable assurance of physical integrity. Existing solutions, including active electronic monitoring systems and passive tamper-evident indicators, suffer from key limitations, including lack of standardization, susceptibility to replication, operational fragility, and the absence of time-identifying detection capabilities.

To address these gaps, we propose Endpoint Assurance, a lightweight, hardware-enabled tamper detection system that combines two complementary physical unclonable functions (PUFs): an optical PUF embedded in tamper-evident tape and a capacitive mesh PUF integrated within the parcel structure. Together, these mechanisms provide both surface-level and structural integrity verification, and an embedded microcontroller continuously records timestamped capacitance data to create a persistent, cryptographically secure record of potential tampering events.

The system is designed to operate in constrained logistics environments without reliance on external power or communication infrastructure, while also mitigating the threat of adversarial replication due to enhanced sophistication of the bi-detection components. By enabling both irreversible physical evidence of tampering and time-resolved detection, Endpoint Assurance overcomes key shortcomings of existing approaches and provides a scalable framework for high-assurance supply chain integrity. This work contributes to a new integration of PUF-based authentication with embedded detection and secure logging, with applications spanning multiple domains that rely on the security of high-impact or high-value cargo.

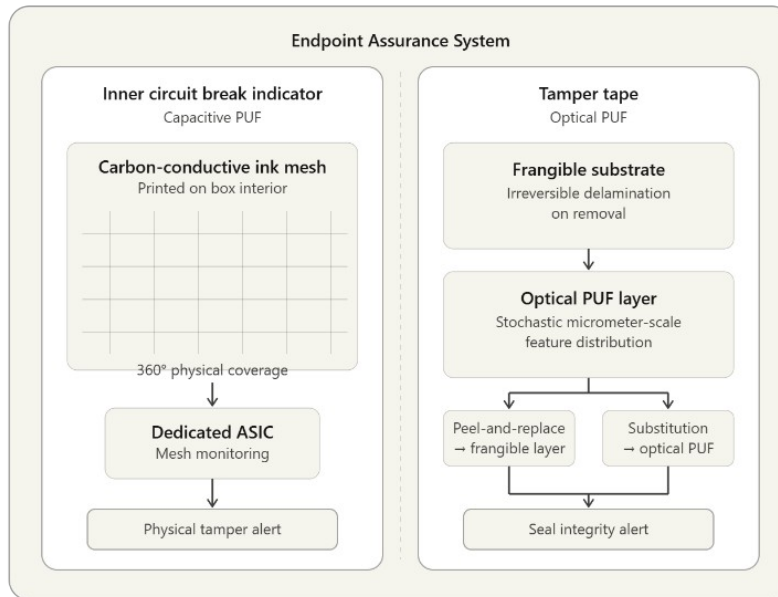


Figure 1: Architecture of the proposed system, comprising two tamper detection components: an inner capacitive PUF (carbon-conductive ink mesh monitored by a dedicated ASIC) and an outer tamper tape optical PUF (substrate with stochastic optical features), limiting peel-and-replace and substitution attacks respectively.

2 Motivation

2.1 Market analysis

Modern supply chains increasingly depend on multi-tier logistics networks in which cargo passes through numerous intermediaries (freight forwarders, transshipment hubs, customs brokers, and third-party warehouses) before reaching its destination. Total seaborne trade reached nearly 12.3 billion metric tons in 2024 (UNCTAD, 2024), and more than 60% of the world's consumer goods are transported via shipping containers, representing nearly \$14 trillion in value (Contimod, 2025). Within that volume, the shipments of greatest concern are not commodity goods but high-value, sensitive, or mission-critical cargo (pharmaceuticals, consumer electronics, semiconductors, precision instruments, luxury goods, and defense components) where the consequences of tampering extend well beyond financial loss.

The scale of the problem across these categories is significant. Counterfeit consumer electronics cost more than \$100 billion per year globally, and bogus computer chips cost U.S. companies an estimated \$7.5 billion annually (Supply Chain Brain, 2021). Annual financial losses from counterfeit electronics exceed \$100 billion in the electronics sector alone, with the impact extending across automotive (\$40 billion), defense (\$32 billion), and healthcare (\$25 billion) (Astute Group, 2025). In pharmaceuticals, exposure is both financial and humanitarian: the counterfeit drug market costs more than \$200 billion per year, with more than one million people annually losing their lives to counterfeit medicines (Supply Chain Brain, 2021). The global tamper-evident packaging market surpassed \$20 billion in 2024, with annual growth exceeding 6% (Eco Security Tape, 2025), itself a proxy measure of how broadly industries now recognize physical integrity as a critical vulnerability.

What ties these categories together is not just their value, but the extended, opaque logistics chains through which they travel. High-value shipments often pass through multiple hands (manufacturers, freight forwarders, customs authorities, warehouses, and last-mile carriers), with each transfer point increasing exposure to tampering, misrouting, or unauthorized access (Iraq Live, 2026). For organizations managing sensitive cargo, this fragmentation creates a fundamental visibility problem: at each handoff, the ability to verify the physical integrity of a shipment degrades, while the incentive for a sophisticated actor to intervene only grows.

2.2 Shortfalls of existing solutions

2.2.1 *Tamper tape difficulty of use*

Active electronic monitoring systems (GPS trackers, data loggers, and smart seals) represent one response to this threat environment. These systems provide continuous telemetry but require power, communication infrastructure, and the ability to attach or retrieve hardware from the shipment. These requirements are often difficult to meet in closed logistics environments, restricted transit zones, or air cargo contexts where regulations limit the use of embedded

electronics. The result is a class of solution that is operationally brittle: effective under ideal conditions, but unreliable precisely in the high-risk, low-visibility transit segments where assurance matters most.

Passive mechanical indicators such as tamper-evident tape offer a lower-friction alternative. They require no power, no infrastructure, and no specialized training to apply. Industries including pharmaceuticals, finance, and electronics are required by law to use tamper-evident packaging for products like prescription drugs, sensitive documents, or high-value chips (Eco Security Tape, 2025), yet regulatory mandates do not resolve the underlying operational challenges. Tape must be applied uniformly across all seams and interfaces, on surfaces that may be contaminated, irregular, or subject to temperature and humidity variation in transit. Visual assessment at the destination is subjective, dependent on inspector training and attention, and typically performed under time pressure in receiving environments not optimized for careful review. The gap between controlled testing performance and operational deployment is substantial.

2.2.2 Tamper tape ease of replication

The more serious challenge is posed by a sophisticated adversary with the resources, tradecraft, and intent to open a package without leaving any detectable trace. The defense and high-value electronics sectors illustrate how far this capability has developed. It has become virtually impossible to detect a counterfeit component simply by looking at it (Retronix, 2025), and the same is true of the packaging that surrounds it. A skilled actor can source matching tape stocks, replicate seal geometries, condition adhesive residue, and reconstruct surface features with enough fidelity to pass visual inspection. Cloning and tampering techniques have become more prevalent, with counterfeiters able to physically remove a silicon die, delayer it completely, and build a functional clone that from an external perspective looks and performs like the authentic device (SMT Corp., 2023). The packaging layer around such a substitution is a far simpler problem.

Against this threat profile, conventional tamper evidence provides little assurance. U.S. Senate Armed Services Committee hearings identified more than 1,800 cases of counterfeit electronic parts and over one million individual suspect parts in the DoD supply chain in just a two-year period (A-CAPP, n.d.), the overwhelming majority of which passed through normal custody and inspection processes undetected. If the indicator itself can be faithfully reproduced, its presence at delivery proves nothing. The problem is not just that tampering occurs; it is that tampering can be made to look like it never did.

2.2.3 Lack of timestamp

Neither active nor passive solutions adequately address the scenario in which an adversary interferes with a shipment during transit while the legitimate owner has no access during that window. The integrity question cannot be answered in real time; it can only be evaluated after the fact, at the point of delivery. In documented defense supply chain cases, DOD was unaware that counterfeit electronic parts had been installed on certain defense systems until after external

investigation, and in one case, suspect parts changed hands five times before reaching the end assembler, with neither DOD nor the prime contractor aware of the original source (U.S. Senate Committee on Armed Services, 2012).

Current tamper-evident approaches are stateless: they record the binary fact of whether a seal appears intact, but provide no information about when a disturbance occurred, at which custody transfer, or under whose possession. Without a persistent, unforgeable record tied to a specific point in time and a specific physical configuration, there is no mechanism to establish whether a package was disturbed before or after any given handoff, leaving the chain of evidence fundamentally incomplete. In high-stakes applications involving defense components, pharmaceutical integrity, or sensitive electronics, this gap is not a minor limitation. It is a structural failure that sophisticated adversaries have already learned to exploit.

3 Concept of Operations (CONOPS)

The Endpoint Assurance solution is a lightweight hardware system that is integrated into a shipping parcel by a (potentially unwitting) sender, with feedback from/to a witting recipient. The system is designed to function for long durations in environments that standard packages may endure.

3.1 Sender-side CONOPS

The sender receives a shipment kit containing a parcel fitted with a discrete integrated capacitive mesh and microcontroller, a roll of tape embedded with infrared (IR)-reflective particles designed to resemble ordinary packing tape, and a commercial off-the-shelf (COTS) IR scanner. To begin, the sender downloads and installs the Endpoint Assurance application on their device (e.g., smartphone) and establishes a secure connection to the microcontroller using a technology such as near-field communication (NFC). The sender then places the sensitive items intended for shipment inside the parcel.

Next, the sender cuts a sufficient length of tape from the provided roll to properly seal the parcel. After closing the parcel, the sender applies the tape along the seam to secure it. Using the provided IR scanner, the sender then scans the applied tape, capturing the unique and irreplicable pattern created by the randomly distributed IR-reflective particles within it. This scanned pattern is transmitted to the receiver through the application, who will later verify it using their own IR scanner upon delivery. The sender places the sealed parcel containing sensitive items, the remaining roll of tape, and the IR scanner together into a larger outer box and ships everything as a single package. Together, the capacitive mesh and IR-reflective tape act as physical unclonable functions (PUF), as the unique “fingerprints” they create cannot be replicated for spoofing. Finally, the sender initiates a packaging event through the app. The microcontroller records both the timestamp of the packaging event and the current capacitance reading of the mesh, and the IR scan is associated to the packaging event. The outer box acts as an extra layer of protection for the inner parcel against efforts to access the sensitive contents via methods such as computed tomography (CT).

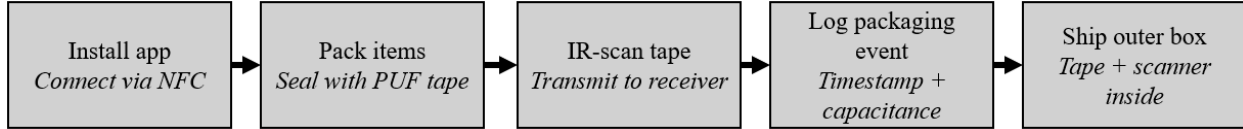


Figure 2: Overview of sender-side CONOPS. The sender receives a package containing IR-reflective tape and an IR scanner to create a unique optical signature for the package, and providing traceability for the packaging event.

3.2 Shipment tampering CONOPS

When properly packaged, both the outer and inner parcels are visually indistinguishable from a standard parcel. If an adversary attempts to open the inner parcel and tamper with its contents, the system leverages the IR-reflective tamper tape and the capacitive mesh to leave permanent evidence of tampering. During transit, the embedded microcontroller records the capacitance of the mesh periodically.

To access the contents of the shipment, an adversary must first open or compromise the outer package. Since the sender is unable to access the roll of tape or IR scanner after closing and sealing the outer box, there are no active security measures to detect tampering on the outer package beyond any physical evidence the adversary leaves behind during entry. However, this does not compromise the overall shipment integrity, as the roll of tape and IR scanner are not considered sensitive items. Instead, the sensitive contents are inside the inner parcel, which is protected by the aforementioned security mechanisms.

If the adversary persists and attempts to access the sensitive contents of the inner parcel, the system's layered detection mechanisms engage. If the tamper tape is damaged through attempts to peel, cut, or otherwise remove it, the embedded IR-reflective particles are permanently altered such that the original pattern cannot be restored. This discrepancy will be identified by the receiver upon scanning. In the event that the adversary bypasses the tape entirely (e.g., by cutting a hole through the side or top of the parcel) the capacitive mesh is physically broken, causing a measurable change in capacitance. This disruption is recorded by the microcontroller and flagged as a potential tampering event.

Any attempt to conceal evidence of tampering within this system is futile. The tape's IR pattern is unique and irreplicable due to microscopic, random, and inherently uncontrollable variations in the distribution of the IR-reflective particles, making it impossible to reproduce even with the most advanced technology. Any physical alterations caused by tampering are therefore permanently embedded in the tape. Although the adversary has access to the remaining roll of IR-reflective tape within the outer package, attempting to reseal the inner parcel with it does not compromise the system's integrity. Any new segment of tape cut from the roll will inherently produce a unique and distinct IR pattern due to the random distribution of the embedded particles. Since the sender transmitted the original tape's scanned pattern to the receiver prior to shipment, any resealing attempt — even with tape from the same roll — will produce a pattern mismatch upon the receiver's verification scan, immediately revealing evidence of tampering.

Similarly, any damage to the capacitive mesh produces significant and detectable fluctuations in capacitance, which are irreversibly recorded by the microcontroller. Physically reconstructing a broken capacitive mesh to restore the original capacitance value is infeasible, as the mesh's conductive pathways are deliberately designed to be fragile and irreparable once severed. Any attempt to bridge or bypass the damaged mesh would introduce measurable inconsistencies in the capacitance readings. Similarly, spoofing or altering the microcontroller's stored data is mitigated through cryptographic protections. All logged data is digitally signed and encrypted, ensuring that any modifications to the records are detectable upon verification. Without access to the cryptographic keys — which are securely stored and never transmitted in plaintext — an adversary cannot fabricate or alter the logged capacitance history without detection.

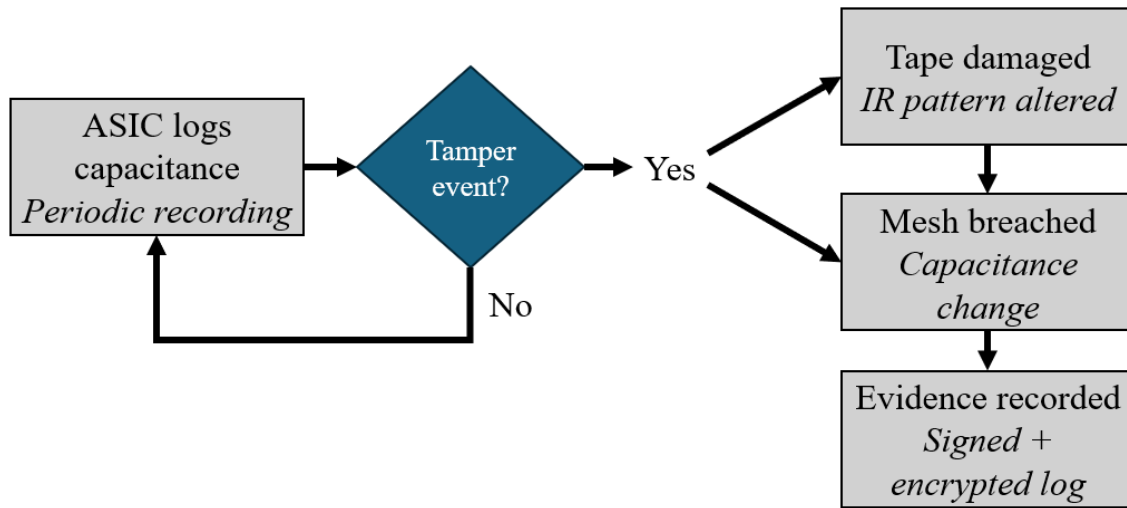


Figure 3: Overview of system CONOPS during transit. The capacitive mesh and IR tape work in tandem to detect if and when tampering occurs.

3.3 Receiver-side CONOPS

Upon delivery, the receiver takes possession of the outer package and begins the verification process. The receiver opens the outer package and retrieves the IR scanner, the remaining roll of tape, and the sealed inner parcel. The receiver first visually inspects both the outer package and the inner parcel for any obvious signs of physical tampering, such as tears, punctures, resealing marks, or other irregularities. While the outer package lacks active security measures, visible damage may serve as an early indicator of a potential compromise.

The receiver then proceeds to verify the integrity of the IR-reflective tamper tape applied to the inner parcel's seam. Using either the packaged or a separate IR scanner, the receiver scans the tape, and the application compares the resulting pattern against the reference pattern originally transmitted by the sender. If the patterns match, the tape has not been disturbed, and the physical seal of the inner parcel is confirmed to be intact. If a mismatch is detected, the system alerts the receiver that the tape has been altered or replaced, indicating a potential tampering event.

Next, the receiver opens the Endpoint Assurance application on their device and establishes a secure connection to the embedded microcontroller via NFC or another supported connection technology. Through the application, the receiver retrieves the full capacitance log recorded during transit. The application compares the recorded capacitance values against the baseline established by the sender at the time of the original packaging event. If all recorded values fall within the predefined tolerance threshold, the mesh is confirmed to be intact and no physical breach of the inner parcel occurred during transit. If any readings deviate beyond the threshold, the application flags the corresponding timestamps as potential tampering events, providing the receiver with a detailed timeline of when the anomaly occurred.

The receiver makes a final determination based on the results of the outlined verification steps. The shipment is considered untampered if the IR tape pattern matches the reference pattern transmitted by the sender, the capacitance readings remain within a predefined tolerance threshold and there are no unexplained gaps in recording intervals, and no visual signs of physical tampering are observed on either the outer or inner parcel. If any of these conditions are not satisfied, it may warrant further action by the receiver to investigate potential tampering activity during the shipping process.

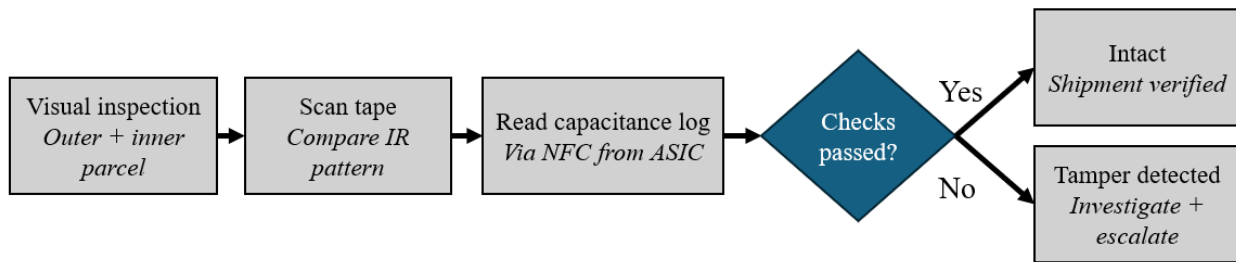


Figure 4: Overview of receiver-side CONOPS. Any tampering that occurs during the shipping process will leave permanent evidence of tampering and can be checked by the receiver.

4 Tamper Indication System

4.1 Technical specifications

The technical specifications detailed in this section represent the requirements in operation for standard use of the Endpoint Assurance solution. The technical specifications take into account functional requirements provided by project sponsors, engineering experience, internal discussions among the development team and capabilities of existing technologies.

The high-level functional specifications for the Endpoint Assurance solution are as follows:

Table 1: Summary of Technical Specifications	
Requirement	Specification
False positive rate: Rate at which solution incorrectly flags a shipment as potentially tampered with despite no actual compromise having taken place.	<1%
Temperature: Survives shipping environments up to ambient temperature.	Min -18°C, Max 71°C
Operational time: Minimum time for which system needs to continuously operate while in use.	180 days
Humidity: Survives shipping environments up to humidity specification.	90% RH
Volumetric footprint: Maximum volume occupied by total system.	8 in ³

The Endpoint Assurance solution is mostly a technology-ready system. Most components are commercial off-the-shelf and therefore fall into a Technology Readiness Level (TRL) of 8 and 9, with the exception of the conductive mesh, which is under active development.

4.2 Physical Unclonable Function (PUF)

A Physical Unclonable Function (PUF) is a hardware-based security primitive that exploits inherent and uncontrollable variations introduced during the manufacturing process to generate unique, device-specific responses (Pappu et al., 2002). These variations arise from random physical phenomena at the micro or nanoscale level, making each PUF instance fundamentally distinct, even among devices produced on the same manufacturing line using identical design specifications (Herder et al., 2014). When presented with an input, known as a challenge, a PUF produces a corresponding output, known as a response, that is determined by its unique physical microstructure. This challenge-response pair (CRP) mechanism forms the basis of PUF-based authentication and identification protocols (Gassend et al., 2002).

The security of a PUF rests on several foundational properties. First, a PUF must be physically unclonable, meaning that no two PUF instances, even those fabricated under identical conditions, should produce the same response to a given challenge due to the inherent randomness of the manufacturing process (Maes, 2013). Second, a PUF must be tamper-evident, such that any physical attempt to probe, reverse-engineer, or extract information from the PUF alters its microstructure and consequently changes its responses, rendering the original challenge-response behavior unrecoverable (Gassend et al., 2002). Third, a PUF should be unpredictable, meaning that knowledge of a subset of challenge-response pairs should not enable an adversary to predict the response to a new, previously unseen challenge (Rührmair et al., 2010).

PUFs can be broadly categorized based on their physical implementation. Silicon-based PUFs, such as arbiter PUFs and ring oscillator PUFs, exploit manufacturing variations in integrated circuit delay paths and frequency characteristics to generate unique responses (Suh & Devadas, 2007). Memory-based PUFs, such as SRAM PUFs, leverage the random power-up states of static memory cells, which are determined by minute transistor-level asymmetries introduced during

fabrication (Holcomb et al., 2009). Optical PUFs, among the earliest proposed, derive their uniqueness from the scattering patterns produced when coherent light passes through a token containing randomly distributed microscopic particles (Pappu et al., 2002). This latter category is particularly relevant to applications involving physical authentication of objects, as the random particle distribution creates a unique and irreplicable optical signature.

The applicability of PUFs extends across a wide range of security domains, including device authentication, cryptographic key generation, anti-counterfeiting, and supply chain integrity verification (Herder et al., 2014). In the context of tamper detection systems, the PUF paradigm can be applied to physical security features, such as coatings, seals, or tapes embedded with randomly distributed particles, where the unique and irreplicable pattern produced by the random physical structure serves as a verifiable fingerprint that is permanently altered by any physical interference (Maes, 2013).

4.3 Tamper tape optical PUF

The tamper tape integrates a frangible substrate with an embedded optical physical unclonable function (PUF) to defeat two complementary attack classes. Peel-and-replace attempts, in which an adversary removes and reapplies the original seal, are caught by the frangible layer, which delaminates irreversibly on removal and destroys the indicator. Substitution attempts, in which an adversary applies a visually similar but fresh seal, are caught by the optical PUF, whose micrometer-scale feature distribution is stochastic at manufacture and therefore cannot be reproduced on a substitute tape (Buchanan et al., 2005; Pappu et al., 2002).

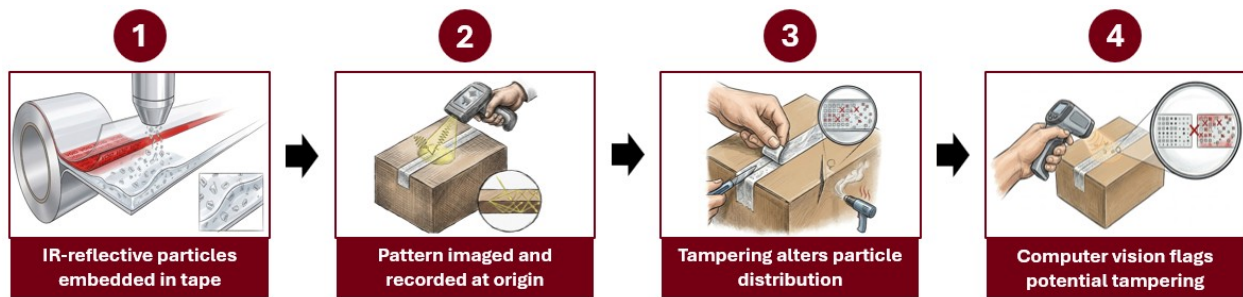


Figure 5: Optical PUF, derived from scalable tamper-evident tape, provides low-cost verification of cargo integrity within existing logistics workflows using embedded IR-reflective particles

4.3.1 Fundamentals

The PUF medium is a stochastic field of ultraviolet-fluorescent particles, commercial thief-detection powder, sandwiched between a frangible kraft-paper substrate and an outer layer of clear packing tape. The kraft layer functions as the tamper-evident base: removing the tape from its adherend requires reversing the adhesive bond, which in this construction delaminates the paper and irreversibly disturbs the embedded particle field. The outer clear layer seals and protects the particles against abrasion, moisture, and handling during transit. Because deposition of the powder is mechanically random at the scale of individual grains, each length of tape carries a unique

particle distribution that cannot be deterministically reproduced, a property first formalized for optical PUFs by Pappu et al. (2002) and demonstrated for packaging-scale surface fingerprints by Buchanan et al. (2005).

Authentication follows the two-scan enrollment-verification protocol standard for optical PUFs (Škorić et al., 2005). At origin, the applied tape is illuminated with a narrow-band ultraviolet source and imaged with a smartphone camera to produce a baseline descriptor; at receipt, the same tape segment is re-imaged under equivalent conditions. Ultraviolet excitation produces strong fluorescence against a nominally dark substrate, maximizing contrast for downstream feature extraction. A discrepancy between the two descriptors indicates either physical disturbance of the seal or substitution of the tape; the modeling layer described in Section 4.3.3 is responsible for distinguishing these events from benign measurement variation and handling wear.

4.3.2 Testing

Characterization of the tape is organized around a paired baseline–verification imaging protocol designed to populate the evaluation metrics reported in Section 4.1. Three pair classes are collected. Untampered re-scan pairs, in which the same tape segment is imaged twice with no intervening disturbance, establish the intrinsic measurement variation attributable to camera noise, registration error, and illumination drift, and set the lower bound on achievable false-reject rate. Tamper pairs apply the two canonical attacks of interest—peel-and-reapply and cut-and-reseal—and quantify the separation between genuine tampering and the untampered distribution. Wear-condition pairs subject the tape to representative environmental and handling stresses (surface abrasion, particulate contamination, and repeated handling) referenced to the distribution-testing profile of ASTM D4169 (ASTM International, 2022), so that any operating threshold can be validated against conditions a qualified shipment is expected to survive without triggering a false alarm.

4.3.3 Modeling Parallel Approaches

To demonstrate the viability of an optical PUF, two complementary modeling paths have been pursued in parallel.

4.3.3.1 Siamese Neural Network Trained on Synthetic Data

A deep metric-learning path, implemented as a Siamese convolutional network (Bromley et al., 1993; Koch et al., 2015), has been trained and pre-validated on synthetic tape imagery, a pixel-grid surrogate of the particle field subjected to randomized perturbations and approximately an order of magnitude of data augmentation, in order to stand up the training and evaluation infrastructure before paired physical samples are available in quantity. The synthetic regime is inherently optimistic: Siamese architectures typically require on the order of 10^3 – 10^4 labeled real pairs to generalize across intra-class nuisance variation such as illumination, pose, and wear (Koch et al., 2015; Shorten & Khoshgoftaar, 2019), and their learned representations resist the per-feature inspection that is desirable for certification in regulated supply-chain contexts (Rudin, 2019).

4.3.3.2 Classical Machine Learning Pipeline

To remedy the problems described above and generate results requiring less data, we developed a fully interpretable classical image-processing pipeline whose component stages are well established in the computer-vision and physical-authentication literature.

Each verification begins with geometric registration and cropping of the tape region in both the baseline and receipt images. A scale-invariant key point detector (Lowe, 2004) is applied to localize individual fluorescent particles and the local speckle substructure they produce, and mutual-nearest-neighbor matching with a ratio test identifies corresponding particles across the two scans. The resulting distribution of matched, displaced, and unmatched key points provides a direct signal of particle-field disruption, which is the expected outcome of peel-and-replace and cut-and-reseal attacks on the tape. In parallel, a Canny edge map (Canny, 1986) is computed on each image and differenced after registration; edges present only in the receipt image are retained as candidate new cuts, creases, or adhesive failures, while edges common to both scans—pre-existing substrate features, printed markings, and benign handling wear—are suppressed. A combined decision rule aggregates the key point-matching score and the normalized residual edge energy, with thresholds calibrated on the paired data described in Section 4.3.2.

The feature vectors from the image analysis are passed into two classifiers—a histogram gradient-boosting model and logistic regression. We evaluate both with Group k-fold cross-validation: the data are split into k folds by group, so for each fold the model trains on some groups and is validated on held-out groups, which avoids training and testing on correlated rows from the same underlying unit. After comparison, the selected model is trained on the entire training set. The classical pipeline yields per-feature evidence that a human inspector can audit, a property that is difficult to recover from end-to-end learned representations (Rudin, 2019).

To train and validate this pipeline, 24 tape samples were prepared. Each one was placed in a dark enclosure and imaged twice before tampering. Then, each one was either carefully peeled or cut down the middle and imaged two more times. 20 of the samples were then used to train the model and finetune the threshold. The validation set was made up of 4 samples, which generated 20 pairs of images—16 tampered and 4 untampered. On this small dataset, the best model tuned to minimize false positives, achieved an accuracy rate of 85%, and a false positive rate of 0%.

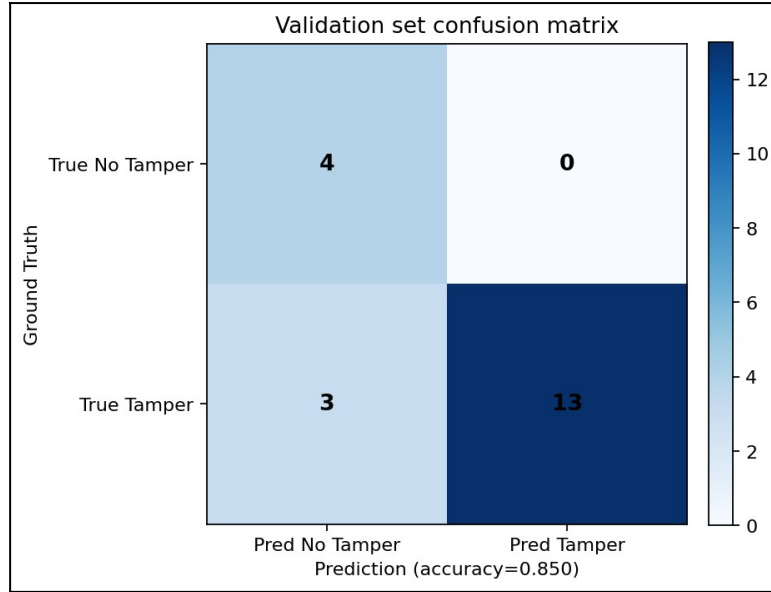


Figure 6: The confusion matrix generated by the best trained classification model.

4.4 Circuit break indicator capacitive PUF

The internal layer of the Endpoint Assurance system relies on a “Capacitive PUF”, consisting of a carbon-conductive ink mesh printed on the inside of a standard cardboard shipping box. Monitored by a dedicated ASIC, this system combines 360° coverage of the package with high sensitivity to adversary physical tampering.

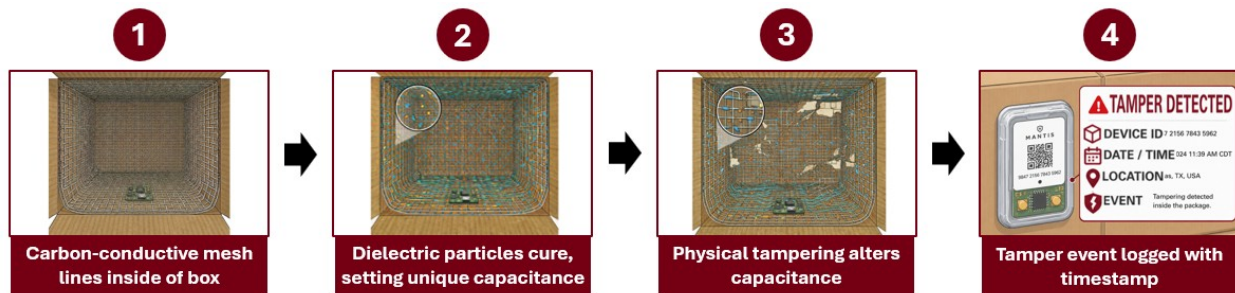


Figure 7: Capacitive PUF is an internal circuit break indicator embedded within the packaging that traces and logs when a package is breached using carbon-based tracing

4.4.1 Fundamentals

A standard parallel-plate capacitor is created when two conductive metal plates are separated by an insulating material (a dielectric). When a voltage difference is applied across the plates, an electric field builds up in the dielectric, storing energy in the form of electrical charge.

In an adaptation of the FORTRESS system developed by the Fraunhofer Institute, a network of such capacitors is created using a mesh of top (TX) and bottom (RX) carbon-conductive electrodes.

Because they are arranged orthogonally, every intersection where a TX electrode crosses over an RX electrode forms a tiny, localized parallel-plate capacitor. Microscopic variations in printing the carbon-conductive ink mesh mean that the exact capacitance at every intersection is slightly different, creating the unique physical “fingerprint” that is the hallmark of a PUF. The system is monitored during transit by a dedicated ASIC, which allows for instantaneous logging of disruptions to the mesh.

Beyond the proof-of-concept, several key design choices are made to enable the system’s smooth function. Variations in absolute capacitance caused by “global” manufacturing variations would normally swamp the unique fingerprint of each node, so an antiphase excitation signal is used to null out the physical physics of the enclosure, leaving only the minuscule femtofarad difference to be digitized. The traces are printed with variable spacing and a sparse center to keep the overall resistor-capacitor baseline low and stable, further improving signal-to-noise ratio and suppressing false positives. The system is also adapted to suit its anti-tampering function by normalizing groups of traces together and zeroing the mean, increasing the overall security of the system by turning any localized physical attack into a mathematically global event for that group.

The unique capacitive fingerprint of each mesh arises from three uncontrollable physical phenomena inherent to the manufacturing process. First, as the dielectric material cures and sets between the electrode layers, it does so randomly, creating micro-variations in the spacing and permittivity between the top and bottom carbon traces that are unique to each individual unit. Second, high-speed printing of carbon-conductive ink onto a porous substrate like cardboard ensures that trace edges are never perfectly straight and trace thickness is never perfectly uniform, so the effective capacitor surface area at each node varies in ways that cannot be replicated. Third, the inherent structural variation of standard cardboard — fiber distribution, minor corrugation warping, and localized differences in ink absorption — further individualizes the capacitance at each intersection. Together these three sources of entropy ensure that even two boxes produced on the same line with identical materials will generate distinct capacitive fingerprints, and that the fingerprint cannot be reproduced even by the original manufacturer.

At the point of sealing, the embedded monitor module performs an enrollment sweep across all rows and columns of the crossbar array, recording the unique baseline capacitance profile generated by these manufacturing variations. This enrolled profile serves as the “birth certificate” of the individual package, stored in onboard memory and used as the reference against which all subsequent measurements during transit are compared. Any physical disturbance that alters the geometry of the array – severing a trace, deforming the dielectric layer, or introducing a conductive probe – produces a measurable and irreversible deviation from this baseline, which the monitor module logs with a timestamp.

4.4.2 Testing

Initial testing is focused on proof-of-concept, both to ensure viability and to control costs. The challenges involved in monitoring femtofarad-level differences mean that testing protocols are initially focused on evaluating the efficacy of the system design itself: rejection of false positives, resilience to external EM influences, durability in high or low temperatures and despite rough handling or vibration, and effectiveness against attempted adversary bypass efforts. A secondary set of tests is focused on developing optimal parameters for the final concept, particularly the spacing and placement of the capacitive traces, and the design and operation of the dedicated ASIC to balance polling rate with battery life.

For further testing beyond the proof of concept, testing of the capacitive PUF is organized around a three-zone threshold model that distinguishes between normal operating variation, environmental drift, and genuine tamper events. Zone A defines the stable operating state, in which capacitance at each node fluctuates within $\pm 3\sigma$ of the enrolled baseline due to inherent electrical noise in the measurement circuit. Zone B captures environmental drift, characterized by slow, proportional, and spatially global shifts in capacitance across the entire grid consistent with changes in ambient temperature or humidity; rather than triggering a tamper alarm, Zone B deviations prompt a dynamic recalibration of the baseline. Zone C defines the tamper condition, in which a sudden, localized, or large-magnitude shift, on the order of greater than 10σ , is detected at one or a small cluster of nodes, inconsistent with the global pattern expected from environmental causes.

The spatial pattern of capacitance change is a key discriminator in this model and is a primary focus of testing. Under normal variation, node values follow a Gaussian distribution across the grid, with minor systematic differences between central and edge nodes attributable to parasitic capacitance. Under environmental conditions, the surface plot of grid capacitance moves up or down as a single sheet. Under a physical attack, the surface plot exhibits a sharp localized discontinuity: a severed trace produces a sudden drop toward zero at the affected node, while contact from a probe or finger produces a spike at a specific coordinate. Testing protocols are therefore designed to characterize all three pattern types under controlled conditions, including thermal cycling across the operational temperature range, humidity exposure, mechanical shock and vibration, and deliberate tamper attempts including trace scratching, panel cutting, and probe attachment, in order to establish defensible zone boundaries and validate the spatial discrimination logic.

4.4.2.1 Prototype testing

To validate the feasibility of the capacitive mesh as a tamper detection mechanism, two prototype arrays were constructed and tested using a 3x3 grid of measurement nodes. Each array used a different conductive material to form the capacitive mesh, allowing for a comparative evaluation of detection sensitivity and material performance under simulated tampering conditions.

The first prototype was constructed using interwoven strips of copper tape arranged in a 3x3 grid, producing nine distinct measurement nodes at the intersections. Baseline capacitance values were recorded across all nine nodes prior to tampering. A simulated tampering event was then performed by physically breaching the mesh at Node 5, the central intersection. Post-tampering capacitance measurements were collected at all nodes and compared against the baseline. The results demonstrated a dramatic response at the tampered node, exhibiting a capacitance change of -2.3 million percent at node 5, effectively indicating a complete loss of measurable capacitance at the breach location. Surrounding nodes showed comparatively minor variations, with nodes 4 and 6 notably showing smaller but significant changes of +43.4 percent and +47.7 percent respectively. The capacitance change results are shown in Table 2.

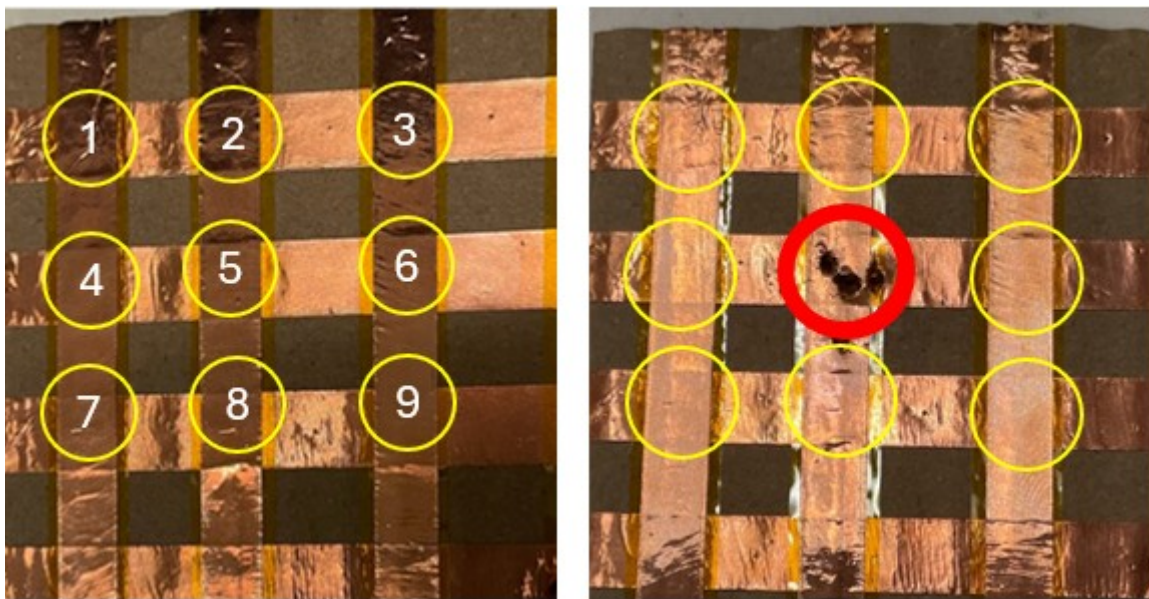


Figure 8: Copper tape array prototype, with labeled untampered nodes (left) and tampered node 5 (right).

The second prototype replaced the copper tape with carbon conductive paint, applied in the same 3x3 grid configuration to produce nine measurement nodes. Baseline capacitance values were again recorded prior to tampering, after which a simulated breach was performed at Nodes 5 and 6.

The carbon conductive paint array also demonstrated detectable changes at the tampered nodes, with Node 6 showing the largest deviation at -51.3 percent. However, the overall response was significantly less pronounced than that of the copper tape array, with several non-tampered nodes also exhibiting notable variation, including Node 4 at -14.9 percent, Node 1 at +5.1 percent, and Node 9 at +42.1 percent. This greater spread of variation across non-tampered nodes suggests that while the carbon paint mesh is capable of detecting tampering, its signal-to-noise ratio is lower than that of the copper tape implementation, potentially increasing the risk of false positives or ambiguous readings under operational conditions.

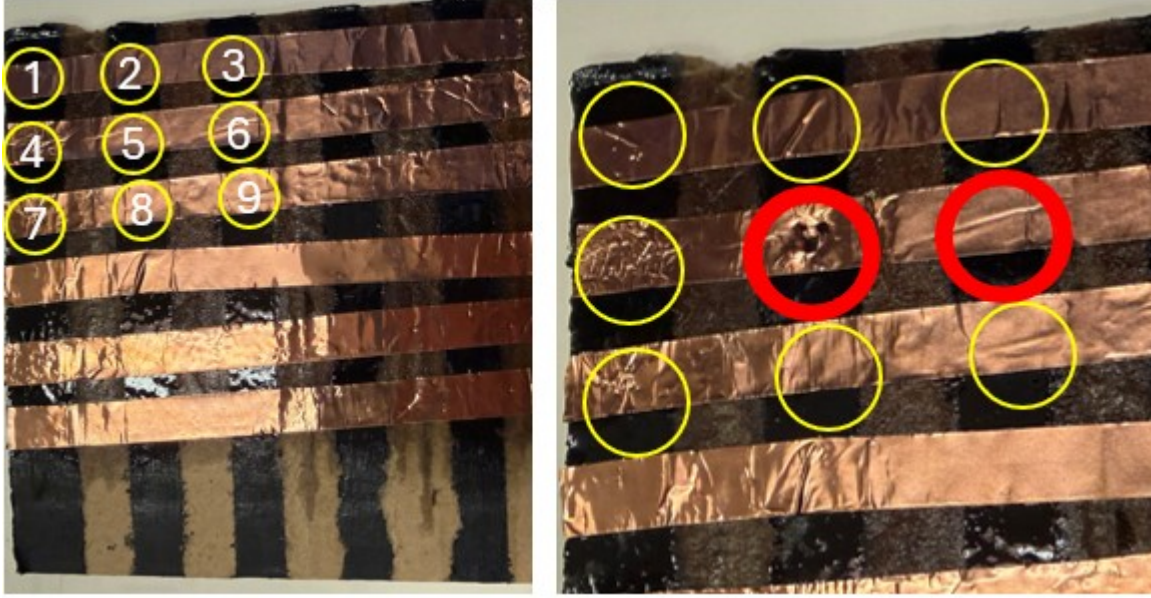


Figure 9: Carbon conductive paint array prototype, with labeled untampered nodes (left) and tampered nodes 5 and 6 (right).

Table 2: Capacitive Mesh Prototypes Testing Results							
Node	Copper Tape	Capacitance	Percent Change	Carbon Conductive Ink	Capacitance	Percent Change	
1		+2.4%			+5.1%		
2		+7.4%			-8.4%		
3		+0.5%			-4.0%		
4		+43.4%			-14.9%		
5		-2,300,000%			-9.1%		
6		+47.7%			-51.3%		
7		+1.5%			-8.4%		
8		+8.8%			-0.8%		
9		+1.6%			+42.1%		

4.4.3 Modeling

Modeling of this solution is used primarily to assess and predict the performance of different trace layouts and densities, and to develop capacitance thresholds for the ASIC to fine-tune sensitivity to false positives and environmental interference.

Beyond the proof-of-concept, modeling of the capacitive PUF addresses both the physical behavior of the mesh and the algorithmic pipeline required to compress raw analog measurements into a stable cryptographic output. The compression pipeline operates in three stages. First, a challenge-response mechanism selects specific node pairs from the crossbar array to query, with the challenge acting as a digital input that determines which row and column intersections are measured. Second, rather than recording absolute capacitance values, which are susceptible to drift from global

environmental changes, the system uses differential sensing, in which a comparator circuit evaluates pairs of adjacent nodes and outputs a binary digit based on their relative magnitudes. Because ambient environmental changes affect the entire array approximately equally, taking the difference between adjacent nodes cancels out global noise and produces a stable binary string that reflects only the unique physical microstructure of that specific mesh. Third, the raw binary string is passed through an error-correcting fuzzy extractor to compensate for residual bit-flip errors introduced by microscopic electrical noise. Following the polar code construction demonstrated by Garb (2023), this stage compresses the noisy raw measurement into a stable cryptographic key, with the Gaussian distribution of femtofarad-range capacitance values across the mesh confirmed by Immler et al. (2019) providing the statistical foundation required for the error-correcting codes to function reliably, and the measurement system architecture described by Obermaier et al. (2018) informing the hardware implementation of the differential sensing stage.

Modeling also informs trace layout and density optimization, with the primary objective of maximizing the uniqueness of inter-device fingerprints, quantified as the hamming distance between the enrolled profiles of different boxes, while maintaining the stability of intra-device measurements across repeated reads of the same box under varying environmental conditions. Simulations of different trace geometries and spacing configurations are used to predict the distribution of node capacitances and identify layouts that maximize entropy while keeping the resistor-capacitor baseline low enough to maintain signal-to-noise ratio and suppress false positives.

4.5 Component list

Components used in construction of the optical PUF Tape are:

Table 3: Optical PUF (Tape) Components	
Material	Purpose
PET or BOPP Film (Polypropylene)	Base film layer for tape substrate
Textured Backing	Microtexture layer used to generate diffraction/scatter patterns
Additives	Modify optical scattering or mechanical behavior of film
Frangible Primers	Creates brittle layer that cracks when tape is peeled
Nickel Shim / Master Pattern	Laser-machined or lithographic master used to emboss micropattern
UV-Curable Resin	Replicates micropattern from shim into film
High-Tack HMPSA	Adhesive layer bonding tape to package
Chemical “Solvent-Swell” Dyes	Chemical tamper indicator that reacts to solvent attack
Thermochromic/Photochromic Inks	Optional covert optical indicator or secondary tamper signal

Components used in construction of the circuit break indicator capacitive PUF are:

Table 4: Capacitive PUF (Circuit Break Indicator) Components	
Material	Purpose
Carbon Conductive Ink	Printing conductive tamper mesh traces on packaging
AD7147 CDC	Capacitive sensing of trace integrity / mesh proximity changes
PET Film with Dielectric	Flexible substrate for printed traces or layered mesh structure
Tamper-Evident Epoxy	Encapsulating electronics and preventing probe access
Microcontroller (low-power dev board)	Logging events and interfacing sensors
Accelerometer (MEMS)	Detect motion or shock events inside the box
Photodiode / Light Sensor	Detect box opening via sudden light exposure
Thin Flexible Copper Tape	Rapid prototyping of mesh or antenna structures
Small LiPo Battery	Power internal sensor node
LiPo Charger IC / Module	Recharge battery during development
Cardboard prototype boxes	Packaging structure

5 Future work

5.1 Optical PUF Tape

Four priorities guide near-term development of the optical PUF tape: collection of field-representative data, a dedicated reader, a migration from ultraviolet to infrared excitation, and a head-to-head benchmark of the classical and Siamese matching approaches.

Bench-scale evaluation will be supplemented by a field data-collection campaign in which sealed packages are routed through commercial ground, air, and international parcel carriers and re-imaged on arrival. This will quantify the operational distribution of abrasion, humidity, and thermal excursion and establish defensible false-accept and false-reject thresholds under realistic transit conditions, with environmental loading referenced to ASTM D4169 (ASTM International, 2022). In parallel, the present capture stack—a smartphone paired with a handheld ultraviolet source—will be replaced by a purpose-built reader providing fixed working distance, controlled geometry, and calibrated narrow-band illumination; the illumination- and camera-variability noise characterized for consumer-grade optical fingerprinting (Clarkson et al., 2009) currently dominates the false-positive budget and is expected to compress substantially under a dedicated reader.

A planned migration from ultraviolet-excited fluorescent particles to infrared-reflective particles is expected to simplify both the reader hardware and the matching model: near-infrared imaging penetrates overlying inks and printed labels that compete with the particle signal in the visible

band (Fischer & Kakoulli, 2006), and narrow-band IR illumination is more easily shielded from ambient light than broad-spectrum UV fluorescence. Finally, once paired real-world data are sufficient for domain adaptation, the classical pipeline of Section 4.2.3 and the synthetic-trained Siamese network will be benchmarked on a common held-out field set (Koch et al., 2015; Shorten & Khoshgoftaar, 2019), with performance reported as equal-error rate and area under the detection-error tradeoff curve alongside inference latency and interpretability.

5.2 Capacitive PUF

A parallel effort is underway to develop a capacitive PUF using conductive ink traces and a dedicated ASIC to measure capacitance variations at circuit overlap points. Future work on this component will be scoped once initial development milestones are reached and the approach has been characterized at the bench level.

Near-term development of the capacitive PUF is focused on four priorities: full characterization of the PoC, validation of the complete PUF algorithmic pipeline, optimization of trace geometry and ASIC parameters, and a transition toward a manufacturable printed architecture. The immediate next step is systematic bench-level characterization of the existing mesh. This involves baseline mapping of every node across repeated measurements to extract the per-node noise floor, followed by controlled temperature and humidity stress testing to correlate environmental conditions with capacitance drift. This data will be used to calibrate the three-zone thresholding model, distinguishing intrinsic variation, environmental drift, and genuine tamper events, as well as to validate the differential spatial analysis approach, in which each node is compared against its neighbors rather than against an absolute threshold, so that global environmental shifts can be distinguished from localized discontinuities characteristic of a physical attack.

Once the characterization dataset is established, work will turn to validating the full PUF algorithmic pipeline. Raw analog capacitance measurements from the crossbar array must be compressed into a stable cryptographic output through a three-stage process: targeted node-pair selection via a digital challenge, conversion of analog measurements to a binary string via differential comparison, and error correction via a fuzzy extractor. Following the polar code construction described by Garb (2023), the goal is to demonstrate that a noisy raw measurement across the array can be reliably reduced to a stable cryptographic key, with the Obermaier et al. (2018) and Immler et al. (2019) results providing the baseline statistical expectations for femtofarad-range capacitance distributions against which our implementation will be benchmarked. The papers mentioned are cited in the main paper that explains the theory behind the PUF, which can be found [here](#).

Parameter optimization will run in parallel, using the modeling framework described in Section 4.4.3 to evaluate the effect of trace spacing, density, and polling rate on the balance between tamper sensitivity, false positive rate, and battery life. Key outputs of this phase will be a finalized trace layout and a validated ASIC configuration for the dedicated capacitance-to-digital converter.

The longer-term scaling objective is to transition from the current hand-assembled PoC to a continuous multi-layer printing process integrated directly into standard corrugated cardboard manufacturing. In this architecture, parallel carbon-conductive traces are printed along the X-axis via high-speed flexographic or screen printing, a dielectric layer is sprayed or laminated over them, and a second pass of carbon ink is printed orthogonally to complete the crossbar matrix. This approach leverages the inherent microscopic chaos of commercial printing, namely the variations in ink deposition, dielectric curing, and substrate topography, as the entropy source for PUF uniqueness, rather than requiring controlled fabrication. Validating that this manufacturing process produces sufficient inter-device uniqueness, measured as hamming distance between boxes, while maintaining intra-device stability across repeated measurements of the same box, will be the primary evaluation milestone for this phase of development.

6 Acknowledgements

The Endpoint Assurance team would like to thank the Engineering Innovation: Global Security Systems teaching staff, namely Dr. AJ Perez, Arjun Chandar, Gene Keselman, Jayna Wadhwa and Liam Fontaine, for their guidance and support throughout the design and development of MANTIS. The team would also like to thank the U.S. Government Agency sponsors for their advice and expertise that played a crucial role in shaping the system.

7 References

- ASTM International. (2022). Standard practice for performance testing of shipping containers and systems (ASTM D4169-22). ASTM International. <https://doi.org/10.1520/D4169-22>
- Astute Group. (2025, December 1). Counterfeit electronics threaten supply chain sustainability across global industries. <https://www.astutegroup.com/news/industrial/counterfeit-electronics-threaten-supply-chain-sustainability-across-global-industries/>
- Bromley, J., Guyon, I., LeCun, Y., Säcker, E., & Shah, R. (1993). Signature verification using a "Siamese" time delay neural network. *Advances in Neural Information Processing Systems*, 6, 737–744.
- Buchanan, J. D. R., Cowburn, R. P., Jausovec, A.-V., Petit, D., Seem, P., Xiong, G., Atkinson, D., Fenton, K., Allwood, D. A., & Bryan, M. T. (2005). 'Fingerprinting' documents and packaging. *Nature*, 436(7050), 475. <https://doi.org/10.1038/436475a>
- Canny, J. (1986). A computational approach to edge detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, PAMI-8(6), 679–698. <https://doi.org/10.1109/TPAMI.1986.4767851>
- Center for Anti-Counterfeiting and Product Protection. (n.d.). Counterfeit parts in the U.S. Department of Defense supply chain. Michigan State University.

<https://a-capp.msu.edu/article/counterfeit-parts-in-the-u-s-department-of-defense-supply-chain/>

Clarkson, W., Weyrich, T., Finkelstein, A., Heninger, N., Halderman, J. A., & Felten, E. W. (2009). Fingerprinting blank paper using commodity scanners. In 2009 30th IEEE Symposium on Security and Privacy (pp. 301–314). IEEE.
<https://doi.org/10.1109/SP.2009.7>

Eco Security Tape. (2025, October 30). How tamper evident technology protects B2B supply chain. <https://ecosecuritytape.com/tamper-evident-technology-guide/>

Fischer, C., & Kakoulli, I. (2006). Multispectral and hyperspectral imaging technologies in conservation: Current research and potential applications. *Studies in Conservation*, 51(sup1), 3–16. <https://doi.org/10.1179/sic.2006.51.Supplement-1.3>

Gassend, B., Clarke, D., van Dijk, M., & Devadas, S. (2002). Silicon physical random functions. *Proceedings of the 9th ACM Conference on Computer and Communications Security*, 148–160. <https://doi.org/10.1145/586110.586132>

Helfmeier, C., Boit, C., Nedospasov, D., & Seifert, J.-P. (2013). Cloning physically unclonable functions. In 2013 IEEE International Symposium on Hardware-Oriented Security and Trust (HOST) (pp. 1–6). IEEE. <https://doi.org/10.1109/HST.2013.6581556>

Herder, C., Yu, M.-D., Koushanfar, F., & Devadas, S. (2014). Physical unclonable functions and applications: A tutorial. *Proceedings of the IEEE*, 102(8), 1126–1141.
<https://doi.org/10.1109/JPROC.2014.2320516>

Holcomb, D. E., Burleson, W. P., & Fu, K. (2009). Power-up SRAM state as an identifying fingerprint and source of true random numbers. *IEEE Transactions on Computers*, 58(9), 1198–1210. <https://doi.org/10.1109/TC.2008.212>

Immler, V., Obermaier, J., König, M., Hiller, M., & Sigl, G. (2019). B-TREPID: Batteryless tamper-resistant envelope with a PUF and integrity detection. In 2019 IEEE International Symposium on Hardware Oriented Security and Trust (HOST) (pp. 49–58). IEEE.
<https://doi.org/10.1109/HST.2019.8741029>

Iraq Live. (2026, February 5). Supply chain risk management: Strategies for high-value shipments. <https://www.iraq-live.com/2026/02/05/supply-chain-risk-management-strategies-for-high-value-shipments/>

Koch, G., Zemel, R., & Salakhutdinov, R. (2015). Siamese neural networks for one-shot image recognition. In *Proceedings of the 32nd International Conference on Machine Learning, Deep Learning Workshop*.

- Lowe, D. G. (2004). Distinctive image features from scale-invariant keypoints. *International Journal of Computer Vision*, 60(2), 91–110.
<https://doi.org/10.1023/B:VISI.0000029664.99615.94>
- Maes, R. (2013). *Physically unclonable functions: Constructions, properties and applications*. Springer. <https://doi.org/10.1007/978-3-642-41395-7>
- Maiti, A., Gunreddy, V., & Schaumont, P. (2013). A systematic method to evaluate and compare the performance of physical unclonable functions. In P. Athanas, D. Pnevmatikatos, & N. Sklavos (Eds.), *Embedded systems design with FPGAs* (pp. 245–267). Springer.
https://doi.org/10.1007/978-1-4614-1362-2_11
- Mordor Intelligence. (2025). Tamper evident packaging market size, share & 2030 growth trends report. <https://www.mordorintelligence.com/industry-reports/tamper-evident-packaging-market>
- Obermaier, J., Immler, V., Hiller, M., & Sigl, G. (2018). A measurement system for capacitive PUF-based security enclosures. In *Proceedings of the 55th ACM/ESDA/IEEE Design Automation Conference (DAC)* (pp. 1–6). ACM.
<https://doi.org/10.1145/3195970.3196090>
- Pappu, R., Recht, B., Taylor, J., & Gershenfeld, N. (2002). Physical one-way functions. *Science*, 297(5589), 2026–2030. <https://doi.org/10.1126/science.1074376>
- Retronix. (2025, April 1). Is your supply chain safe? Combatting the threat of counterfeit electronic components. <https://retronix.com/combating-counterfeit-threats/>
- Rudin, C. (2019). Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence*, 1(5), 206–215.
<https://doi.org/10.1038/s42256-019-0048-x>
- Rührmair, U., Sehnke, F., Sölter, J., Dror, G., Devadas, S., & Schmidhuber, J. (2010). Modeling attacks on physical unclonable functions. *Proceedings of the 17th ACM Conference on Computer and Communications Security*, 237–249.
<https://doi.org/10.1145/1866307.1866335>
- Shorten, C., & Khoshgoftaar, T. M. (2019). A survey on image data augmentation for deep learning. *Journal of Big Data*, 6(1), Article 60. <https://doi.org/10.1186/s40537-019-0197-0>
- Škorić, B., Tuyls, P., & Oprea, W. (2005). Robust key extraction from physical uncloneable functions. In J. Ioannidis, A. Keromytis, & M. Yung (Eds.), *Applied cryptography and network security (ACNS 2005)* (LNCS 3531, pp. 407–422). Springer.
https://doi.org/10.1007/11496137_28

- SMT Corp. (2023). Counterfeit electronic parts in the critical infrastructure supply chain. <https://smtcorp.com/wp-content/uploads/2023/05/Growing-Threat-of-Counterfeit-Electronic-Parts-in-the-Critical-Infrastructure-Supply-Chain-Releasable-1.pdf>
- Suh, G. E., & Devadas, S. (2007). Physical unclonable functions for device authentication and secret key generation. *Proceedings of the 44th Annual Design Automation Conference*, 9–14. <https://doi.org/10.1145/1278480.1278484>
- Supply Chain Brain. (2021, August 24). Combatting supply chain counterfeiting with technology. <https://www.supplychainbrain.com/blogs/1-think-tank/post/33575-combatting-supply-chain-counterfeiting-with-technology>
- Tuyls, P., Schrijen, G.-J., Škorić, B., van Geloven, J., Verhaegh, N., & Wolters, R. (2006). Read-proof hardware from protective coatings. In L. Goubin & M. Matsui (Eds.), *Cryptographic hardware and embedded systems — CHES 2006* (LNCS 4249, pp. 369–383). Springer. https://doi.org/10.1007/11894063_29
- United Nations Conference on Trade and Development. (2024, October 22). High freight rates strain global supply chains, threaten vulnerable economies. <https://unctad.org/news/high-freight-rates-strain-global-supply-chains-threaten-vulnerable-economies>
- U.S. Senate Committee on Armed Services. (2012, May 21). Senate Armed Services Committee releases report on counterfeit electronic parts. <https://www.armed-services.senate.gov/press-releases/senate-armed-services-committee-releases-report-on-counterfeit-electronic-parts>